DSAA3073: Theories in Data Science

Tutorial 2A: From One Reliable Estimate to a Reliable Learner

150-minute core tutorial · 15-minute break · 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

Hoeffding controls one hypothesis chosen before the data. What must change when the data choose the hypothesis?

A learning algorithm does not merely estimate the error of a rule fixed in advance. It looks at a sample, compares many candidate rules, and returns one of them. That final act of selection is where a correct concentration calculation can become an incorrect learning argument.

We will build the statistical learning problem before asking for a guarantee. One binary classification example carries the notation from population risk to empirical risk, then to empirical risk minimization and realizable PAC learning. The last block returns to Hoeffding and repairs data-dependent selection with a simultaneous event over a finite class.

The three core blocks take **45, 50, and 55 minutes**. Take the scheduled 15-minute break after the first block. The last 60 minutes are protected for questions, slower reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0–45	Declare the learning objects; compare population and empirical risk	45 min core
45–60	Scheduled break	15 min
60–110	Select by empirical risk; state the realizable PAC contract	50 min core
110–165	Diagnose selection; derive the finite-class guarantees	55 min core
165–225	Questions, recovery, alternative calculations	60 min buffer

Name the problem before measuring success

Consider a credit-approval system. An input x records an applicant's features, and the label $y \in \{0,1\}$ records whether approval is judged safe. A rule h maps an input to a predicted label. We could immediately count training mistakes, but that number has no stable meaning until we say which rules are allowed, how mistakes are measured, and how the data are generated.

Definition: Statistical learning framework

A batch statistical learning problem declares all of the following:

- an input domain $\mathcal{X}$ and a label space $\mathcal{Y}$;
- a **hypothesis class** $\mathcal{H}$ of candidate maps $h : \mathcal{X} \to \mathcal{Y}$;
- a loss $\ell(h(x), y)$ that assigns a numerical cost to a prediction;
- a data distribution $\mathcal{D}$ on labeled examples (x, y); and
- an ordered i.i.d. training sample $S = ((x_i, y_i))_{i=1}^{m} \sim \mathcal{D}^m$.

Here i.i.d. means that the m labeled examples are independent and each has the same distribution $\mathcal{D}$.

For the approval example, take $\mathcal{Y} = \{0,1\}$ and use the zero-one loss

$$\ell(h(x), y) = \mathbf{1}_{\{h(x) \neq y\}}, \tag{1}$$

which equals 1 for a mistake and 0 otherwise. The distribution $\mathcal{D}$ describes the population from which both training applicants and future applicants are drawn. It need not be known to the learner. The class $\mathcal{H}$ is a modeling choice: it may contain four hand-written rules, all decision trees of a prescribed depth, or another declared family.

IN-CLASS CHECK

Prompt: A team supplies 200 labeled applicants and a neural-network training program, but does not state a loss or the population from which the applicants were sampled. Has it specified the statistical learning problem above? Name what is observed and what is still missing.

Hint: Match the description against the six declared objects.

Answer:

> No. The 200 labeled pairs and the training program are observed, but the loss and data distribution are missing; the allowed hypothesis class may also be ambiguous unless the architecture and parameter restrictions are declared. A list of candidate predictors alone is not a complete learning problem.

The error we want and the error we can compute

Fix one hypothesis $h \in \mathcal{H}$. Its performance on the population is an expectation over a fresh labeled example.

Definition: Population risk

For the declared distribution and loss, the **population risk** of a fixed hypothesis h is

$$R(h) = E_{(X,Y)\sim\mathcal{D}}[\ell(h(X), Y)]. \tag{2}$$

With zero-one loss, $R(h) = P_{(X,Y)\sim\mathcal{D}}(h(X) \neq Y)$, the probability that h misclassifies a fresh example.

The canonical symbol is $R(h)$. It contains no sample subscript because it is a property of h, $\mathcal{D}$, and the loss, not of one observed dataset.

Definition: Empirical risk

On the observed sample $S = ((x_i, y_i))_{i=1}^m$, the **empirical risk** is

$$\hat{R}_{S(h)} = \frac{1}{m}\sum_{i=1}^{m} \ell(h(x_i), y_i). \tag{3}$$

It is the average loss on the training sample.

The hat and the subscript S matter. Dropping them changes the quantity from a sample average into a population expectation.

Worked example: The same rule under two averages

Suppose a stylized population has four applicant types with probabilities $0.40, 0.25, 0.20, 0.15$. A fixed rule h_0 is correct on types 1 and 3 and wrong on types 2 and 4. Under zero-one loss,

$$R(h_0) = 0.25 + 0.15 = 0.40. \tag{4}$$

Now evaluate the same h_0 on six observed labeled applicants. If its predictions are $(1, 0, 0, 1, 1, 0)$ while the labels are $(1, 1, 0, 1, 0, 0)$, it makes two mistakes:

$$\hat{R}_{S(h_0)} = \frac{2}{6} = \frac{1}{3}. \tag{5}$$

The two values differ without contradiction. They average the same loss for the same hypothesis over different objects: $\mathcal{D}$ versus the realized sample S.

Shared object	Population branch	Sample branch
h and ℓ	fresh $(X, Y) \sim \mathcal{D}$	$S = ((x_i, y_i))_{i=1}^m$
average	$E_{\mathcal{D}}[\ell(h(X), Y)]$	$\frac{1}{m}\sum_i \ell(h(x_i), y_i)$
name	$R(h)$	$\hat{R}_{S(h)}$
directly observed?	No, unless $\mathcal{D}$ is fully known	Yes, after S is observed

Table 1: One hypothesis and one loss feed two averages. The orange column is distributional; the blue column is computed from the observed sample.

Trace the first row before reading on: the hypothesis and loss do not change between columns. Only the object being averaged changes. This is why a small training error is evidence about, but not identical to, small population risk.

IN-CLASS CHECK

Prompt: For the six-example calculation above, identify the random object before sampling, the observed object after sampling, and the quantity that remains distributional. Then write both risk formulas without looking back.

Hint: The sample is random before it is drawn. The population risk is not replaced by a training average.

Answer:

Before sampling, $S \sim \mathcal{D}^6$ is random. After sampling, the six labeled pairs and $\hat{R}_{S(h_0)} = \frac{1}{3}$ are observed. The population quantity $R(h_0) = E_{\mathcal{D}}[\ell(h_0(X), Y)]$ remains distributional. The required formulas are $R(h) = E_{\mathcal{D}}[\ell(h(X), Y)]$ and $\hat{R}_{S(h)} = m^{-1} \sum_{i=1}^{m} \ell(h(x_i), y_i)$.

The sample branch gives us numbers for every candidate rule. That creates an obvious selection procedure—and a less obvious statistical danger.

Select a rule, then state the promise

Suppose $\mathcal{H} = \{h_1, h_2, h_3, h_4\}$ and the observed empirical risks are

h_1	h_2	h_3	h_4
$\frac{2}{6}$	$\frac{1}{6}$	0	$\frac{1}{6}$

The green entry is the smallest. Selecting h_3 uses the sample twice: first to compute four risks, then to choose an index.

Definition: Empirical risk minimization

An **empirical risk minimizer** is any

$$\hat{h}_S \in \arg\min_{h\in\mathcal{H}} \hat{R}_{S(h)}. \tag{6}$$

Empirical risk minimization (ERM) returns such a hypothesis. If an exact minimum is unavailable, the learner must declare its optimization tolerance relative to the infimum.

ERM minimizes the quantity we can compute. It does not directly minimize $R(h)$, which depends on the unknown distribution.

IN-CLASS CHECK

Prompt: If h_2 and h_4 tie for the smallest empirical risk, is ERM undefined? If an oracle instead returns the hypothesis with smallest population risk, is that ERM?

Hint: Read the set-valued $\arg\min$ literally.

Answer:

> A tie does not make ERM undefined: either minimizer may be returned, or the algorithm may use a declared tie-breaking rule. An oracle that minimizes population risk is not performing ERM, because ERM minimizes $\hat{R}_{S(h)}$, the sample quantity.

A guarantee with its quantifiers visible

We first study the **realizable** setting. A distribution is realizable by $\mathcal{H}$ when there exists $h^\star \in \mathcal{H}$ with $R(h^\star) = 0$. For zero-one loss, some allowed hypothesis predicts the label correctly with probability one. This is an assumption about the pair $(\mathcal{D}, \mathcal{H})$, not a conclusion from observing zero training error.

Definition: Realizable PAC learnability

A hypothesis class $\mathcal{H}$ is **realizably PAC learnable** if there exist one learning algorithm A and a sample-complexity function $m_{\mathcal{H}} : (0,1)^2 \to N$ such that, for every $(\varepsilon, \delta) \in (0,1)^2$, for every distribution $\mathcal{D}$ realizable by $\mathcal{H}$, and for every sample size $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$P_{S\sim\mathcal{D}^m}(R(A(S)) \leq \varepsilon) \geq 1 - \delta. \tag{7}$$

If A is randomized, the probability also includes its internal randomness. PAC abbreviates "probably approximately correct."

The two parameters play different roles:

- ε is the **accuracy tolerance**: success means population risk at most ε;
- δ is the **failure probability**: confidence is at least $1 - \delta$.

The order matters. The same learner A must work for every allowed (ε, δ) and every realizable distribution once the sample is large enough. The probability is over the random sample—not over a new test point inside $R(A(S))$, because that fresh-example randomness has already been averaged in the definition of population risk.

Choose	$\to$	Draw	$\to$	Run	$\to$	Wrap in probability
$(\varepsilon, \delta), \mathcal{D}, m$	$\to$	$S \sim \mathcal{D}^m$	$\to$	$A(S)$	$\to$	$R(A(S)) \leq \varepsilon$ with probability $\geq 1 - \delta$

Table 2: The universal choices enter on the left; the sample draw creates the probability space; the success event concerns the returned hypothesis.

Worked example: Realizable and non-realizable are distributional claims

Let $\mathcal{X} = \mathbb{R}$ and let $\mathcal{H}$ be threshold rules $h_{t(x)} = \mathbf{1}_{\{x \geq t\}}$ for a declared finite set of thresholds.

In the first population, labels are deterministically generated by one threshold $t^\star$ included in the class. Then $R(h_{t^\star}) = 0$, so the problem is realizable.

In the second population, the same feature value $x = 0$ receives label 0 with positive probability and label 1 with positive probability. Every deterministic threshold makes an error on one of those cases with positive probability. No $h \in \mathcal{H}$ has zero population risk, so this distribution is not realizable by the class—even if a small sample happens to be fitted perfectly.

IN-CLASS CHECK

Prompt: Put these clauses in a valid order: (i) draw $S \sim \mathcal{D}^m$; (ii) choose any realizable $\mathcal{D}$ and any (ε, δ); (iii) require $R(A(S)) \le \varepsilon$; (iv) take $m \ge m_{\mathcal{H}}(\varepsilon, \delta)$; (v) assign probability at least $1 - \delta$. Which object is random?

Hint: The sample-size threshold is chosen before the sample is drawn.

Answer:

> The order is (ii), (iv), (i), run A, then (iii) with (v): for every (ε, δ) and every realizable $\mathcal{D}$, every $m \ge m_{\mathcal{H}}(\varepsilon, \delta)$ satisfies $P_{S \sim \mathcal{D}^m}(R(A(S)) \le \varepsilon) \ge 1 - \delta$. The sample S is random; if A randomizes, its internal randomness is included too. The distribution is universally quantified, not sampled from a distribution over distributions.

Worked example: Reading a sample-complexity certificate

Suppose a theorem for a particular class certifies $m_{\mathcal{H}}(0.08, 0.05) = 1200$. The statement means that every realizable distribution for that class, with at least 1200 i.i.d. training examples, yields a returned hypothesis with population risk at most 0.08 with probability at least 0.95. It does not say that 1200 is necessary, that every realized sample succeeds, or that the learner runs quickly.

Samples are not running time

A sample-complexity theorem counts examples. A running-time theorem counts computational operations under a declared representation. These are separate claims. Even when a finite class admits a small sample bound, a literal ERM procedure that scans an enormous encoded class may be computationally infeasible.

IN-CLASS CHECK

Prompt: A proof shows that a class is PAC learnable with a sample size polynomial in $\frac{1}{\varepsilon}$ and $\log(\frac{1}{\delta})$. The proposed learner finds an ERM hypothesis by enumerating all 2^{2^d} encoded candidates. Which conclusion is justified: statistical learnability, efficient learnability, both, or neither?

Hint: Do not turn a statement about examples into one about operations.

Answer:

> The sample statement supports statistical learnability, assuming the other PAC conditions hold. It does not support efficient learnability. Exhaustive enumeration of 2^{2^d} candidates is not polynomial in the representation parameter d. A separate algorithmic argument would be required.

The PAC contract tells us what success means. It does not yet justify applying Hoeffding to $A(S)$ or $\hat{h}_S$, because both are chosen after looking at S.

Protect a choice made after seeing the data

Return to one hypothesis h fixed before observing S. Under zero-one loss, the random variables

$$Z_i = \ell(h(X_i), Y_i) \in [0, 1] \tag{8}$$

are i.i.d., with $E[Z_i] = R(h)$ and $m^{-1} \sum_i Z_i = \hat{R}_{S(h)}$. The Chapter 1 Hoeffding result therefore gives, for every $\gamma > 0$,

Result: Fixed-hypothesis deviation

If h is fixed independently of the i.i.d. sample S, then

$$P\left(\left|\hat{R}_{S(h)} - R(h)\right| > \gamma\right) \le 2 \exp(-2m\gamma^2). \tag{9}$$

The bounded loss supplies the range $[0, 1]$; independence of the examples supplies the product structure required by Hoeffding.

The condition "h is fixed independently of S" is active. ERM instead returns $\hat{h}_S$, an index selected from the same empirical risks that appear in the deviation.

The smallest training error can be the luckiest error

Imagine eight hypotheses evaluated on the same sample of size 20. In a pure-noise population, suppose every one of them has population risk $\frac{1}{2}$. The table shows one possible realized set of empirical risks.

hypothesis	h_1	h_2	h_3	h_4	h_5	h_6	h_7	h_8
$\hat{R}_{S(h)}$	$\frac{11}{20}$	$\frac{9}{20}$	$\frac{10}{20}$	$\frac{7}{20}$	$\frac{11}{20}$	$\frac{8}{20}$	$\frac{5}{20}$	$\frac{10}{20}$

selected after seeing S $j(S) = 7$	$\rightarrow$	**fixed-index event family** $E_{h_j}(\gamma) = \left\{\left|\hat{R}_{S(h_j)} - R(h_j)\right| > \gamma\right\}$ for $j = 1, \dots, 8$	$\rightarrow$	**one envelope before selection** $\cup_{j=1}^{8} E_{h_j}(\gamma)$ Outside it, every row is accurate.

Table 3: ERM highlights h_7 only after all rows are evaluated. The linked event family keeps each hypothesis fixed, then places all eight possible failures inside one envelope.

This is not a claim that the displayed outcome is typical. It isolates the logical problem. The selected index

$$j(S) \in \arg\min_{1 \le j \le 8} \hat{R}_{S(h_j)} \tag{10}$$

depends on S. If we write the fixed-h Hoeffding statement and then replace h by $h_{j(S)}$, we have changed an assumption without justification.

IN-CLASS CHECK

Prompt: In the linked diagram, circle the index that depends on S. Then write the fixed-index event $E_{h_j}(\gamma)$, the family $\left\{E_{h_j}(\gamma) : j = 1, \dots, 8\right\}$, and the union that must be controlled. Why is "Hoeffding holds for every fixed h_j, so it holds for the selected $h_{j(S)}$" incomplete?

Hint: Hold j fixed while defining each event. Only the circled $j(S)$ is data-dependent.

Answer:

Circle $j(S)$. For each fixed $j \in \{1, \dots, 8\}$, write

$$E_{h_j}(\gamma) = \left\{\left|\hat{R}_{S(h_j)} - R(h_j)\right| > \gamma\right\}, \tag{11}$$

so the requested family is $\left\{E_{h_j}(\gamma) : j = 1, \dots, 8\right\}$ and its union is $\cup_{j=1}^{8} E_{h_j}(\gamma)$. Each fixed event satisfies the same bounded-loss, i.i.d. Hoeffding assumptions. The selected index $j(S)$, however, is computed from the sample and may search across these events. Outside their union, every one of the eight comparisons is accurate, so the conclusion remains valid for whichever row ERM selects.

One event that covers every finite candidate

Let $\mathcal{H}$ be finite. For each fixed $h \in \mathcal{H}$, define the bad event

$$E_{h(\gamma)} = \left\{\left|\hat{R}_{S(h)} - R(h)\right| > \gamma\right\}. \tag{12}$$

The definition is indexed by h, but every $E_{h(\gamma)}$ is evaluated on the same random sample. Hoeffding gives $P\left(E_{h(\gamma)}\right) \le 2\exp(-2m\gamma^2)$ for each fixed h. Now place all of those bad outcomes inside one event:

$$P\left(\exists h \in \mathcal{H} : E_{h(\gamma)}\right) = P\left(\cup_{h \in \mathcal{H}} E_{h(\gamma)}\right) \le \sum_{h \in \mathcal{H}} P\left(E_{h(\gamma)}\right) \le 2|\mathcal{H}| \exp(-2m\gamma^2). \tag{13}$$

No independence between the events E_h is needed for the union bound. We only need the class to be finite so that the displayed sum has $|\mathcal{H}|$ terms.

Theorem: Finite-class simultaneous deviation bound

Let $\mathcal{H}$ be a finite hypothesis class, let $S \sim \mathcal{D}^m$ be i.i.d., and let the loss take values in $[0, 1]$. For every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\forall h \in \mathcal{H}, \quad \left|\hat{R}_{S(h)} - R(h)\right| \le \sqrt{\frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2m}}. \tag{14}$$

Equivalently, to make every deviation at most γ, it is sufficient that

$$m \ge \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\gamma^2}. \tag{15}$$

The bound depends on the class through $\ln|\mathcal{H}|$, not $|\mathcal{H}|$ itself. The cardinality appears as a multiplicative factor in the failure probability, and taking logarithms converts that multiplication into addition.

Worked example: Why the simultaneous event protects ERM

Let $\hat{h}_S$ minimize empirical risk and let $h^\star$ minimize population risk in the finite class. On the event where every hypothesis has deviation at most γ,

$$R\left(\hat{h}_S\right) \le \hat{R}_{S\left(\hat{h}_S\right)} + \gamma \le \hat{R}_{S(h^\star)} + \gamma \le R(h^\star) + 2\gamma. \tag{16}$$

The middle inequality is ERM. The first and last inequalities use the same simultaneous event for two hypotheses, one selected from data and one chosen by population risk. A fixed-hypothesis event for only one row cannot justify this chain.

IN-CLASS CHECK

Prompt: Let $|\mathcal{H}| = 1024$, $\delta = 0.05$, and target simultaneous deviation $\gamma = 0.10$. Compute a sufficient m. What excess-risk conclusion does the ERM chain give?

Hint: Use $m \ge \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\gamma^2}$ and round upward.

Answer:

Here

$$m \ge \frac{\ln\left(2 \cdot \frac{1024}{0.05}\right)}{2 \cdot 0.1^2} = \frac{\ln(40960)}{0.02} \approx 531.02. \tag{17}$$

Thus $m = 532$ is sufficient. On the simultaneous event, the displayed ERM chain gives $R\left(\hat{h}_S\right) \le R(h^\star) + 0.20$. The deviation tolerance is $\gamma = 0.10$, while the excess-risk term is $2\gamma = 0.20$; these must not be conflated.

A sharper argument when a consistent target exists

The two-sided square-root bound applies even when no hypothesis has zero risk. Under realizability and zero-one loss, a consistent ERM admits a different, sharper proof. Call a hypothesis **consistent with S** when $\hat{R}_{S(h)} = 0$.

Fix a hypothesis with $R(h) > \varepsilon$. Its probability of making no mistake on m independent examples is

$$P(h \text{ is consistent with } S) = (1 - R(h))^m \le (1 - \varepsilon)^m \le \exp(-m\varepsilon). \tag{18}$$

If the problem is realizable, an ERM can return a consistent hypothesis because the zero-risk $h^\star$ has zero empirical risk with probability one. Failure means that at least one bad hypothesis with population risk above ε is consistent. A union bound gives

$$P\left(\exists h \in \mathcal{H} : R(h) > \varepsilon \text{ and } \hat{R}_{S(h)} = 0\right) \le |\mathcal{H}| \exp(-m\varepsilon). \tag{19}$$

Theorem: Finite-class realizable PAC bound

For a finite hypothesis class with zero-one loss, assume realizability and let ERM return a consistent hypothesis. For every $(\varepsilon, \delta) \in (0,1)^2$, it is sufficient that

$$m \geq \frac{1}{\varepsilon}\left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right) \tag{20}$$

to guarantee $P\left(R\left(\hat{h}_S\right) \leq \varepsilon\right) \geq 1 - \delta$.

This is the realizable $\frac{1}{\varepsilon}$ rate. It is not the same formula as the two-sided simultaneous-deviation rate, which scales as $\frac{1}{\gamma^2}$. The two proofs answer different questions and use different assumptions.

IN-CLASS CHECK

Prompt: Again take $|\mathcal{H}| = 1024$ and $\delta = 0.05$, now with realizable PAC accuracy $\varepsilon = 0.10$. Compute the consistent-realizable sample bound. Compare it carefully with the previous $m = 532$ calculation.

Hint: Use natural logarithms and round upward. Do not call γ and ε the same guarantee.

Answer:

The realizable calculation is

$$m \geq \frac{\ln 1024 + \ln 20}{0.10} \approx \frac{6.9315 + 2.9957}{0.10} \approx 99.27, \tag{21}$$

so $m = 100$ is sufficient. It directly guarantees population risk at most 0.10 for a consistent ERM under realizability. The earlier $m = 532$ bound guaranteed simultaneous absolute deviation at most 0.10 and therefore only an ERM excess-risk term of 0.20. The assumptions and conclusions differ, so the numbers are not interchangeable.

Where the logarithm comes from

For the simultaneous bound, requiring $2|\mathcal{H}|\exp(-2m\gamma^2) \leq \delta$ gives

$$-2m\gamma^2 \leq \ln\delta - \ln 2 - \ln|\mathcal{H}|, \tag{22}$$

and hence

$$m \geq \frac{\ln 2 + \ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)}{2\gamma^2}. \tag{23}$$

The price of comparing more hypotheses is logarithmic because a finite union creates a multiplicative cardinality factor before we solve the exponential tail inequality for m.

IN-CLASS CHECK

Prompt: With $\gamma = 0.05$, $\delta = 0.05$, and one million hypotheses, compute a sufficient sample size for the two-sided simultaneous event. Then state what changes if the class size is squared.

Hint: Use $\ln\left(2 \cdot \frac{10^6}{0.05}\right) = \ln(4 \cdot 10^7)$.

Answer:

$$m \geq \frac{\ln(4 \cdot 10^7)}{2 \cdot 0.05^2} \approx \frac{17.5044}{0.005} \approx 3500.9, \tag{24}$$

so $m = 3501$ is sufficient. Replacing $|\mathcal{H}|$ by $|\mathcal{H}|^2$ doubles only the $\ln|\mathcal{H}|$ contribution; it does not square the required sample size. Confidence and tolerance terms remain.

Standard language to carry forward

Standard term or notation	Meaning here	Formula or contrast
Statistical learning framework	Domain, labels, class, loss, distribution, and i.i.d. sample	$S = ((x_i, y_i))_{i=1}^{m} \sim \mathcal{D}^m$
Hypothesis class $\mathcal{H}$	Declared candidate maps $h : \mathcal{X} \to \mathcal{Y}$	Plain H is not used for the class
Population risk $R(h)$	Expected loss on a fresh example	$E_{\mathcal{D}}[\ell(h(X), Y)]$
Empirical risk $\hat{R}_{S(h)}$	Average loss on the observed sample	$m^{-1} \sum_i \ell(h(x_i), y_i)$
Empirical risk minimization	Choose a sample-risk minimizer	$\hat{h}_S \in \arg\min_{h\in\mathcal{H}} \hat{R}_{S(h)}$
Realizable PAC learnability	One learner succeeds for every realizable distribution at accuracy ε and confidence $1-\delta$	$P(R(A(S)) \leq \varepsilon) \geq 1-\delta$
Finite-class generalization bounds	Union before selection: simultaneous deviations for bounded loss; direct bad-consistent control for realizable zero-one loss	$\frac{1}{\gamma^2}$ versus $\frac{1}{\varepsilon}$, with $\ln\lvert\mathcal{H}\rvert$ in both

What Tutorial 2B may assume

The following is one reconstruction in two panels, not a list to memorize. Work from the blank declarations before comparing with the answers.

IN-CLASS CHECK

Prompt: Panel 1 — declare and decode. Use a finite set $\mathcal{T}$ of thresholds on a pre-agreed applicant score. Declare $\mathcal{X}, \mathcal{Y}, \mathcal{H}, \ell, \mathcal{D}$ and $S = ((x_i, y_i))_{i=1}^m$. Then write and decode $R(h)$, $\hat{R}_{S(h)}$, the ERM rule, realizability, and the realizable PAC quantifiers using (ε, δ).

Hint: Say what is fixed before sampling, what is unknown, what is random, and what is selected from data.

Answer:

Let $\mathcal{X} = \mathbb{R}$ be the declared score domain and $\mathcal{Y} = \{0, 1\}$, where 1 means "safe to approve." For a finite threshold set $\mathcal{T} \subset \mathbb{R}$, take

$$\mathcal{H} = \{h_t : t \in \mathcal{T}\}, \quad h_{t(x)} = \mathbf{1}_{\{x \geq t\}}, \quad \ell\Big(h_{t(x)}, y\Big) = \mathbf{1}_{\{h_{t(x)} \neq y\}}. \tag{25}$$

The domain, labels, finite candidate class, and zero-one loss are modeling choices fixed before sampling. Let $\mathcal{D}$ be the fixed but unknown population distribution on $\mathbb{R} \times \{0, 1\}$, and let $S = ((x_i, y_i))_{i=1}^m \sim \mathcal{D}^m$ be the random i.i.d. training sample, observed after it is drawn.

For $h \in \mathcal{H}$,

$$R(h) = E_{(X,Y)\sim\mathcal{D}}[\ell(h(X), Y)] \tag{26}$$

is its expected loss on a fresh population draw, whereas

$$\hat{R}_{S(h)} = \frac{1}{m} \sum_{i=1}^{m} \ell(h(x_i), y_i) \tag{27}$$

is the observable average loss on this sample. ERM is the data-dependent choice

$$\hat{h}_S \in \arg\min_{h \in \mathcal{H}} \hat{R}_{S(h)}, \tag{28}$$

so $\hat{h}_S$, unlike any fixed h_t, depends on S. Realizability means $\exists h^* \in \mathcal{H} : R(h^*) = 0$. The realizable PAC contract is: there exist a learner A and a function $m_{\mathcal{H}} : (0, 1)^2 \to N$ such that, for every $(\varepsilon, \delta) \in (0, 1)^2$, every realizable $\mathcal{D}$, and every $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$P_{S\sim\mathcal{D}^m}(R(A(S)) \leq \varepsilon) \geq 1 - \delta. \tag{29}$$

IN-CLASS CHECK

Prompt: Panel 2 — reconstruct both finite-class routes. First use bounded loss and two-sided deviations at tolerance γ. Then specialize to zero-one loss, realizability, and a consistent ERM at target risk ε. For each route, write the event family, union probability, sample bound, conditions, and purpose. Finally, identify the step that is generally useless for infinite $\mathcal{H}$.

Hint: The two-sided route controls every empirical-to-population comparison; the realizable route controls hypotheses that are bad yet make no training errors.

Answer:

Two-sided simultaneous route. Assume $\mathcal{H}$ is finite, the sample is i.i.d., and $\ell \in [0,1]$. For fixed h, let $E_{h(\gamma)} = \left\{\left|\hat{R}_{S(h)} - R(h)\right| > \gamma\right\}$. Then

$$P\left(\cup_{h\in\mathcal{H}} E_{h(\gamma)}\right) \le 2|\mathcal{H}| \exp(-2m\gamma^2). \tag{30}$$

Thus

$$m \ge \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\gamma^2} \tag{31}$$

makes every deviation at most γ with probability at least $1-\delta$. This is the route that protects a data-selected ERM and yields $R\left(\hat{h}_S\right) \le \min_{h\in\mathcal{H}} R(h) + 2\gamma$; its tolerance cost is $\frac{1}{\gamma^2}$.

Consistent-realizable route. Now assume zero-one loss, finite $\mathcal{H}$, realizability, and a learner returning a consistent hypothesis, so $\hat{R}_{S(A(S))} = 0$. For fixed h, define

$$B_{h(\varepsilon)} = \left\{R(h) > \varepsilon \ \text{ and } \ \hat{R}_{S(h)} = 0\right\}. \tag{32}$$

A hypothesis with $R(h) > \varepsilon$ is consistent with probability at most $(1-\varepsilon)^m \le \exp(-m\varepsilon)$, hence

$$P\left(\cup_{h\in\mathcal{H}} B_{h(\varepsilon)}\right) \le |\mathcal{H}| \exp(-m\varepsilon). \tag{33}$$

Therefore

$$m \ge \frac{\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)}{\varepsilon} \tag{34}$$

directly guarantees $R(A(S)) \le \varepsilon$ with probability at least $1-\delta$. Its faster $\frac{1}{\varepsilon}$ rate depends on realizability and consistency; it is not the bounded-loss simultaneous-deviation claim.

In both routes, the finite union produces the factor $|\mathcal{H}|$. Replacing that finite sum by an unrestricted infinite sum is generally useless. Tutorial 2B will ask how many distinct labeling behaviors an infinite class can produce on a finite sample. That sentence is a preview only; no new counting object is assumed here.

Key Takeaway

A fixed-hypothesis guarantee survives data-dependent selection only after it has been strengthened to cover every candidate that selection might return. For a finite class, a union bound does that job in two different ways: a bounded-loss simultaneous-deviation route with $\frac{1}{\gamma^2}$ tolerance cost, or a consistent-realizable zero-one route with $\frac{1}{\varepsilon}$ risk cost. Both turn class cardinality into a logarithmic sample penalty. Tutorial 2B may now rely on the full statistical framework, both risk quantities, ERM, the realizable PAC quantifiers, both finite-class arguments, and all five canonical notation forms without re-teaching their basic use.

Source notes

The official Week 2 Learning Sheet v4 is used to preserve course scope and sequence only: `w02-pac-framework`, printed/PDF pp. 1–7, and `w02-finite-uniform`, printed/PDF pp. 8–13. Definitions, claims, derivations, and examples above were reconstructed from the original textbooks:

- Shalev-Shwartz and Ben-David, *Understanding Machine Learning*, Sections 2.1–2.3 (printed/PDF pp. 33–40) and Section 3.1 (pp. 42–44), for the learning objects, empirical-risk minimization, realizability, and PAC quantifiers;
- Mohri, Rostamizadeh, and Talwalkar, *Foundations of Machine Learning*, 2nd ed., Section 2.1 (printed pp. 9–14; PDF pp. 26–31), for risk and the PAC framework;
- Mohri et al., Sections 2.2–2.3 (printed pp. 15–21; PDF pp. 32–38), especially Corollaries 2.10–2.11 and Theorem 2.13, for the consistent-realizable rate, fixed-hypothesis Hoeffding step, selection warning, and finite-class bound;
- Shalev-Shwartz and Ben-David, Chapter 8, especially printed/PDF pp. 108–110, for the separation between sample sufficiency and computational efficiency.

No claim about shattering, VC dimension, or an arbitrary-frequency sine class is used in this Tutorial.

DSAA3073: Theories in Data Science

Tutorial 2B: Count Finite-Sample Behaviors

150-minute core tutorial · 15-minute break · 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-28

Explorative projection

LESSON BACKBONE

A hypothesis class may be infinite. How many different labelings can it actually realize on a finite sample, and when is that enough for learning?

Tutorial 2A protected a data-dependent choice by controlling every member of a finite class. An infinite class defeats that literal union, but its members can still behave identically on the points we have observed. The relevant question is therefore not how many hypotheses exist in total. It is how many distinct binary label vectors they can produce on a finite set.

We will count those vectors, use intervals on the real line to distinguish a constructive VC lower bound from a universal upper bound, and then prove the Sauer-Shelah lemma by removing one point. The final block records the realizable-PAC consequence of finite VC dimension and marks exactly which stronger equivalence still belongs to Chapter 3.

The three core blocks take **60, 55, and 35 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, slower reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-60	Count finite-sample labelings; prove the VC dimension of intervals	60 min core
60-75	Scheduled break	15 min
75-130	Prove Sauer-Shelah by projection and Pascal's identity	55 min core
130-165	State the realizable-PAC consequence and the Chapter 3 boundary	35 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

What survives when the class is infinite?

Tutorial 2A ended with a finite family of bad events, one for each hypothesis. For an infinite $\mathcal{H}$, simply replacing the finite sum by "a sum over every h" gives no useful bound. Yet infinitely many hypotheses can collapse to the same predictions on a fixed finite set.

Take three distinct ordered points $C = \{x_1, x_2, x_3\}$ with $x_1 < x_2 < x_3$, and threshold rules

$$h_{t(x)} = \mathbf{1}_{\{x \geq t\}}, \quad t \in \mathbb{R}. \tag{35}$$

Moving t across the line changes the label vector only four times:

threshold location	$h_{t(x_1)}$	$h_{t(x_2)}$	$h_{t(x_3)}$	label vector
$t > x_3$	0	0	0	$(0, 0, 0)$
$x_2 < t \leq x_3$	0	0	1	$(0, 0, 1)$
$x_1 < t \leq x_2$	0	1	1	$(0, 1, 1)$
$t \leq x_1$	1	1	1	$(1, 1, 1)$

There are infinitely many choices of t, but only four distinct restrictions to C. A **restriction** records predictions on the chosen points and forgets how the hypothesis behaves elsewhere:

$$\mathcal{H}|_C = \{(h(x_1), ..., h(x_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m. \tag{36}$$

The order merely gives coordinates to the vector. Reordering the same set permutes coordinates and does not change the number of realized labelings.

Definition: Growth function

For a binary hypothesis class $\mathcal{H}$, its **growth function** is

$$\Pi_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X} : |C| = m} |\mathcal{H}|_C|. \tag{37}$$

It is the largest number of distinct binary label vectors that $\mathcal{H}$ can realize on any set of m distinct domain points.

The maximum matters. One inconvenient point configuration may admit few labelings even when another configuration of the same size admits many. The count is also a count of distinct restrictions, not of parameter values or hypotheses. Always

$$1 \leq \Pi_{\mathcal{H}}(m) \leq 2^m \tag{38}$$

for a nonempty binary class. The upper bound is simply the number of all binary vectors of length m.

IN-CLASS CHECK

Prompt: On three ordered points, suppose a class contains 200 hypotheses: 80 induce $(0,0,0)$, 50 induce $(0,0,1)$, 40 induce $(0,1,1)$, and 30 induce $(1,1,1)$. What is $|\mathcal{H}|_C|$? What, if anything, does this prove about $\Pi_{\mathcal{H}}(3)$?

Hint: Collapse duplicate prediction vectors before counting; then read the maximum in the definition.

Answer:

The restriction contains four distinct vectors, so $|\mathcal{H}|_C| = 4$. This proves only $\Pi_{\mathcal{H}}(3) \geq 4$, because the growth function maximizes over all three-point sets. We cannot conclude equality until every three-point set is bounded by four.

The largest possible count, 2^m, is qualitatively different: on that set the class has no missing binary behavior.

Definition: Shattering

A binary hypothesis class $\mathcal{H}$ **shatters** a finite set C when

$$\mathcal{H}|_C = \{0,1\}^{|C|}, \tag{39}$$

equivalently when $|\mathcal{H}|_C| = 2^{|C|}$. Every binary labeling of the chosen points must be realized by some hypothesis in the class.

Realizing many labelings is not enough. If seven of the eight vectors on a three-point set occur, the missing eighth vector prevents shattering.

Intervals: one construction and one obstruction

Let $\mathcal{I}$ be the class of interval classifiers on the real line,

$$h_{a,b}(x) = \mathbf{1}_{\{a \leq x \leq b\}}, \quad a \leq b, \tag{40}$$

with label 1 inside the interval. Fix two points $x_1 < x_2$. Each row below specifies an interval placement; the interval may be moved slightly so that no endpoint coincides with a data point.

target	interval placement	realized?	why
$(0,0)$	away from both points	yes	neither point lies inside
$(1,0)$	small interval around x_1	yes	only x_1 lies inside
$(0,1)$	small interval around x_2	yes	only x_2 lies inside
$(1,1)$	one interval covering both	yes	both points lie inside

Table 4: All four binary vectors on two ordered points are realized. This is the constructive half of the VC calculation.

Now take an **arbitrary** three-point set, written in increasing order $z_1 < z_2 < z_3$. The vector $(1,0,1)$ is impossible: an interval containing both z_1 and z_3 contains every real number between them, including z_2.

z_1 target 1	$\rightarrow$	z_2 target 0	$\rightarrow$	z_3 target 1

Any interval covering the two green endpoints also covers the red middle point.

Table 5: The upper-bound triple is arbitrary. One universally impossible labeling is enough to show that no three-point set is shattered.

Read the two figures with different quantifiers. The first needs one chosen two-point set and constructs every labeling. The second begins with every three-point set and supplies a missing labeling for each one.

IN-CLASS CHECK

Prompt: Complete the two-point construction by choosing explicit closed intervals for $x_1 = 0$ and $x_2 = 2$. Then explain in one quantified sentence why the $(1,0,1)$ argument is stronger than showing that one particular triple fails.

Hint: For the all-zero row, place the interval away from both points. The upper bound must cover an arbitrary ordered triple.

Answer:

One valid choice is $[3,4]$ for $(0,0)$, $[-\frac{1}{2},\frac{1}{2}]$ for $(1,0)$, $[\frac{3}{2},\frac{5}{2}]$ for $(0,1)$, and $[-1,3]$ for $(1,1)$. Thus the chosen set $\{0,2\}$ is shattered. For the upper bound: for every three distinct real points $z_1 < z_2 < z_3$, no interval realizes $(1,0,1)$, so no three-point subset of $\mathbb{R}$ is shattered by $\mathcal{I}$.

Definition: Vapnik-Chervonenkis dimension

The **VC dimension** of a binary hypothesis class $\mathcal{H}$ is

$$\mathrm{VCdim}(\mathcal{H}) = \max\{|C| : C \subset \mathcal{X} \ \text{ and } \ \mathcal{H} \ \text{ shatters } \ C\}. \tag{41}$$

If sets of arbitrarily large finite size can be shattered, then $\mathrm{VCdim}(\mathcal{H}) = \infty$.

For intervals, the two-point construction proves $\mathrm{VCdim}(\mathcal{I}) \geq 2$. The arbitrary-triple obstruction proves $\mathrm{VCdim}(\mathcal{I}) < 3$. Therefore

$$\mathrm{VCdim}(\mathcal{I}) = 2. \tag{42}$$

Result: The two proof jobs in a VC calculation

To establish $\mathrm{VCdim}(\mathcal{H}) = d$:

- **lower bound:** exhibit one d-point set and realize all 2^d labelings;
- **upper bound:** prove that every $(d+1)$-point set has at least one unrealizable labeling.

Shattering one d-point set does not claim that every d-point set is shattered. Failing on one $(d+1)$-point set does not prove the upper bound.

IN-CLASS CHECK

Prompt: A class shatters the vertices of one triangle in $\mathbb{R}^2$ but fails to shatter four collinear points. Which VC conclusion is valid: $\mathrm{VCdim}(\mathcal{H}) = 3$, $\mathrm{VCdim}(\mathcal{H}) \geq 3$, or $\mathrm{VCdim}(\mathcal{H}) < 4$? What would finish equality?

Hint: Separate the existential lower-bound witness from the universal upper-bound obligation.

Answer:

> Only $\mathrm{VCdim}(\mathcal{H}) \geq 3$ follows. Failure on one four-point set is not a universal upper bound. To prove equality, one must show that no four-point set anywhere in the domain is shattered by the class.

The interval example now gives us a finite number $d = 2$. It does not yet say how many labelings intervals - or any other class of VC dimension d - can realize when m is much larger than d. The exact answer begins with a one-point projection.

From a shattering limit to polynomial growth

Result: Sauer-Shelah lemma

Let $\mathcal{H}$ be a binary hypothesis class with $\mathrm{VCdim}(\mathcal{H}) \le d < \infty$, where d is a nonnegative integer. Then for every nonnegative integer m,

$$\Pi_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i}, \tag{43}$$

with $\binom{m}{i} = 0$ for $i > m$. For fixed finite d, the right-hand side is polynomial in m, whereas the unrestricted number of binary labelings is 2^m.

The lemma bounds finite-sample behavior, not the cardinality of $\mathcal{H}$. The class itself may remain infinite. For example, with $m = 5$ and $d = 2$,

$$\sum_{i=0}^{2} \binom{5}{i} = 1 + 5 + 10 = 16, \quad \text{while} \quad 2^5 = 32. \tag{44}$$

For intervals, the exact number of label vectors on five ordered points is $1 + 5 + 4 + 3 + 2 + 1 = 16$: choose one consecutive nonempty block of positive points, or choose the all-zero vector. Here the binomial bound is attained.

IN-CLASS CHECK

Prompt: Compute the exact Sauer-Shelah sum for $m = 6, d = 2$ and compare it with 2^6. Then explain why writing only $O(m^2)$ would lose information needed for this check.

Hint: Use $\binom{6}{0} + \binom{6}{1} + \binom{6}{2}$.

Answer:

> The exact sum is $1 + 6 + 15 = 22$, while $2^6 = 64$. The notation $O(m^2)$ records the fixed-d growth order but hides the exact finite-m value and constants, so it cannot certify the numerical bound 22.

Remove one point and sort the extensions

The proof tracks the largest possible number of label vectors. Let $F(m, d)$ denote the maximum of $|\mathcal{H}|_C|$ over all binary classes of VC dimension at most d and all m-point sets C. This is local proof notation; the course-wide growth-function notation remains $\Pi_{\mathcal{H}}(m)$.

Fix $C = \{x_1, \ldots, x_{m-1}, x_m\}$ and a realized-vector family $V = \mathcal{H}|_C$. Remove the last coordinate from every vector. Let

- U be the set of distinct prefixes on $x_1, \ldots, x_{m-1}$;
- $D \subseteq U$ be the prefixes that occur in V with **both** last labels, $(u, 0)$ and $(u, 1)$.

Every prefix in the set difference $U - D$ contributes exactly one vector to V, while every prefix in D contributes two. Hence

$$|V| = |U - D| + 2|D| = |U| + |D|. \tag{45}$$

For intervals on three ordered points, the realized vectors are

$$V = \{000, 001, 010, 011, 100, 110, 111\}. \tag{46}$$

prefix u	**$(u, 0)$ occurs?**	**$(u, 1)$ occurs?**	**group**
00	yes: 000	yes: 001	D
01	yes: 010	yes: 011	D
10	yes: 100	no	$U - D$
11	yes: 110	yes: 111	D

Table 6: Projection removes only the last coordinate. Four prefixes form U; three have both extensions and form D. Thus $|V| = 4 + 3 = 7$.

The table is not merely bookkeeping. It isolates exactly where the VC dimension must drop.

Why the double-extension family loses one dimension

The prefix family U is a restriction of the original class, so it has VC dimension at most d and therefore $|U| \le F(m - 1, d)$.

The crucial claim is that D has VC dimension at most $d - 1$. Suppose instead that D shattered some d of the first $m - 1$ points. Pick any labeling of those d points and either desired label for x_m. Because the corresponding prefix lies in D, both extensions at x_m occur in V. Thus all 2^{d+1} labelings of those d points together with x_m would occur in V. The original class would shatter $d + 1$ points, contradicting $\mathrm{VCdim}(\mathcal{H}) \le d$.

Therefore $|D| \le F(m - 1, d - 1)$, and the projection gives the recurrence

$$F(m, d) \le F(m - 1, d) + F(m - 1, d - 1). \tag{47}$$

IN-CLASS CHECK

Prompt: In the interval projection table, $D = \{00, 01, 11\}$ on two old coordinates. Show directly that D has VC dimension one: give a shattered one-coordinate set and identify the missing vector that prevents the two-coordinate set from being shattered. Then connect this to $d - 1$ when the original interval class has $d = 2$.

Hint: Project D onto either coordinate for the lower bound; inspect its four possible two-bit vectors for the upper bound.

Answer:

> On the first coordinate, D realizes both labels: 00 or 01 gives label 0, and 11 gives label 1, so one point is shattered. On both coordinates, vector 10 is missing, so the two-point set is not shattered. Hence $\mathrm{VCdim}(D) = 1 = 2 - 1$. This is the concrete instance of the general double-extension argument.

IN-CLASS CHECK

Prompt: Fill the logical gap: if D shattered d old points, why are labelings with last coordinate 0 and labelings with last coordinate 1 both available for every labeling of those old points?

Hint: Use the definition of membership in D, not merely membership in U.

Answer:

Shattering by D means that for each binary labeling u on those d old points, some prefix in D realizes u. Membership in D means that this prefix occurs in the original vector family with both extensions, $(u, 0)$ and $(u, 1)$. Therefore every labeling on the d old points can be paired with either label on x_m, which shatters the $d + 1$ point union.

Pascal's identity closes the induction

We prove by induction on m that

$$F(m, d) \leq \sum_{i=0}^{d} \binom{m}{i}. \tag{48}$$

The boundary cases are visible. With no points, there is one empty labeling, so $F(0, d) = 1$. When $d = 0$, no point can receive both labels, so every restriction family has at most one vector and $F(m, 0) = 1$.

For $m \geq 1$ and $d \geq 1$, apply the recurrence and the induction hypothesis:

$$\begin{aligned} F(m, d) &\leq F(m-1, d) + F(m-1, d-1) \\ &\leq \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\ &= \sum_{i=0}^{d} \left(\binom{m-1}{i} + \binom{m-1}{i-1} \right) \\ &= \sum_{i=0}^{d} \binom{m}{i}. \end{aligned} \tag{49}$$

In the third line, set $\binom{m-1}{-1} = 0$. The last equality is Pascal's identity. Since every particular $\mathcal{H}|_C$ is bounded by $F(m, d)$, taking the maximum over C gives the stated bound on $\Pi_{\mathcal{H}}(m)$.

Worked example: One recurrence entry

For $m = 5, d = 2$, the recurrence and the already established smaller cases give

$$F(5, 2) \leq F(4, 2) + F(4, 1) \leq (1 + 4 + 6) + (1 + 4) = 16. \tag{50}$$

Pascal combines the two sums into $1 + 5 + 10 = 16$. This matches the earlier interval count on five ordered points.

IN-CLASS CHECK

Prompt: Reconstruct the four indispensable proof moves without copying formulas: (i) define the projected families, (ii) count the original vectors, (iii) justify the dimension drop, and (iv) close with a binomial identity.

Hint: Each sentence should state a reason, not only an equation.

Answer:

(i) Remove one coordinate, let U contain all distinct prefixes, and let D contain prefixes with both extensions. (ii) A single-extension prefix contributes one vector and a double-extension prefix contributes two, so $|V| = |U| + |D|$. (iii) If D shattered d old points, both last labels would extend every old labeling, so the original class would shatter $d + 1$ points; hence $\mathrm{VCdim}(D) \leq d - 1$. (iv) Induction gives the two smaller binomial sums, and Pascal's identity combines them into $\sum_{i=0}^{d} \binom{m}{i}$.

Sauer-Shelah has replaced the impossible cardinality $|\mathcal{H}|$ by a finite sample behavior count. One link remains before this combinatorial fact becomes a statistical guarantee; stating that link honestly is the final task.

What finite VC dimension buys - and what it does not yet prove here

Recall Tutorial 2A's realizable setting: binary zero-one loss, a distribution for which some $h^\star \in \mathcal{H}$ has $R(h^\star) = 0$, and an ERM that can return a sample-consistent hypothesis. For a finite class, the direct bad-consistent union bound produced the sample penalty $\ln|\mathcal{H}|$.

Finite VC dimension controls a different finite object. On m points, the class realizes at most $\Pi_{\mathcal{H}}(m)$ distinct label vectors, and Sauer-Shelah bounds that number by a polynomial when $d = \mathrm{VCdim}(\mathcal{H}) < \infty$. A standard realizable VC analysis turns this effective-size control into the following forward implication.

Result: Finite VC dimension implies realizable PAC learnability

For binary classification with zero-one loss, if $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, then $\mathcal{H}$ is distribution-free realizably PAC learnable. A standard ERM upper bound has the order

$$m_{\mathcal{H}}(\varepsilon, \delta) = O\left(\frac{d \ln\left(\frac{1}{\varepsilon}\right) + \ln\left(\frac{1}{\delta}\right)}{\varepsilon}\right). \tag{51}$$

The $\frac{1}{\varepsilon}$ dependence is the realizable rate. This statement does not assert computationally efficient ERM.

The proof map is now precise:

finite VC dimension	$\to$	**Sauer-Shelah**	$\to$	**polynomial restrictions**	$\to$	**realizable PAC upper bound**
$d < \infty$	$\to$	$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}$	$\to$	$\Pi_{\mathcal{H}}(m) = O(m^d)$ for fixed d	$\to$	$O\left(\frac{d \ln(\frac{1}{\varepsilon}) + \ln(\frac{1}{\delta})}{\varepsilon}\right)$

Table 7: Solid arrows record the Chapter 2 forward chain. The middle arrow is the complete combinatorial proof above; the final statistical conversion is the standard realizable VC result stated with its conditions.

There is a subtle warning. The set of restrictions on the observed sample depends on that same random sample, so one may not blindly substitute $\Pi_{\mathcal{H}}(m)$ for $|\mathcal{H}|$ inside Tutorial 2A's fixed finite union and declare the proof finished. A complete generalization argument must handle that dependence. Its stronger, agnostic form is developed in Chapter 3; we do not use that future machinery here.

IN-CLASS CHECK

Prompt: Let $d = 10$, $\varepsilon = 0.05$, and $\delta = 0.01$. Without inventing the hidden absolute constant, compute the expression inside the realizable big-O bound and identify the roles of its two terms.

Hint: Use natural logarithms. Keep the answer as a scale calculation, not an exact sample certificate.

Answer:

The displayed scale is

$$\frac{10\ln 20 + \ln 100}{0.05} \approx \frac{29.957 + 4.605}{0.05} \approx 691.2. \tag{52}$$

The $d\ln(\frac{1}{\varepsilon})$ term charges for combinatorial complexity at the requested accuracy, while $\ln(\frac{1}{\delta})$ charges for confidence. Because the theorem is stated with big-O, 691.2 is not itself a certified sufficient sample size; an absolute constant is suppressed.

Keep the proved chain separate from the future equivalence

ERM under realizability	→	**finite VC dimension and growth control**	→	**realizable PAC forward guarantee**

Chapter 3, not used here: remove realizability, establish the needed generalization control, and prove the full agnostic learnability equivalence. Symmetrization is named there and derived there.

Table 8: Solid boxes and arrows are current knowledge. The dashed orange box marks a future proof obligation, not a theorem already established in this Tutorial.

The distinction is not cosmetic. Chapter 2 has established the combinatorial measure, its exact growth theorem, and the realizable forward consequence. It has not proved that finite VC dimension is equivalent to every agnostic or uniform-convergence formulation, nor has it derived a general agnostic rate.

IN-CLASS CHECK

Prompt: Mark each claim C (established in Chapter 2) or N (needs Chapter 3): (a) intervals have VC dimension two; (b) finite VC dimension yields polynomial growth; (c) finite VC dimension implies realizable PAC learnability; (d) finite VC dimension is equivalent to agnostic ERM learnability; (e) a symmetrization derivation. Correct any overclaim.

Hint: Use the solid-versus-dashed proof map.

Answer:

> Claims (a), (b), and the forward implication (c) are C. Claims (d) and (e) are N. The current statement is narrower: under binary zero-one loss and realizability, finite VC dimension gives a distribution-free PAC guarantee. The agnostic equivalence and its proof require Chapter 3's generalization machinery.

Standard language to carry forward

Standard term or notation	**Meaning here**	**Formula or proof cue**
Restriction $\mathcal{H}\vert_C$	Distinct label vectors induced on a fixed finite set	Count vectors, not hypotheses
Growth function $\Pi_{\mathcal{H}}(m)$	Largest restriction count over all m-point sets	$\max_{\vert C\vert=m}\vert\mathcal{H}\vert_C\vert$
Shattering	Every binary labeling of one fixed set is realized	$\mathcal{H}\vert_C = \{0,1\}^{\vert C\vert}$
VC dimension $\mathrm{VCdim}(\mathcal{H})$	Largest shattered-set size, or infinity	one lower-bound witness + universal upper bound
Sauer-Shelah lemma	Finite VC dimension forces polynomial finite-sample growth	$\Pi_{\mathcal{H}}(m) \le \sum_{i=0}^{d}\binom{m}{i}$
Finite-VC realizable PAC learnability	For binary zero-one loss, finite VC dimension gives the forward realizable guarantee	$O\left(\frac{d\ln(\frac{1}{\varepsilon})+\ln(\frac{1}{\delta})}{\varepsilon}\right)$

Reconstruct the Chapter 3 entry

IN-CLASS CHECK

Prompt: Starting only from $\mathcal{H}$ and a finite set C, reconstruct the chain from restrictions to a realizable PAC conclusion. Your answer must define $\Pi_{\mathcal{H}}(m)$, shattering, and $\mathrm{VCdim}(\mathcal{H})$; state Sauer-Shelah with conditions; give the two proof jobs for a VC equality; and mark the exact point where Chapter 3 begins.

Hint: Do not replace the quantified definitions by slogans such as “model complexity.”

Answer:

On $C = \{x_1, ..., x_m\}$, the restriction $\mathcal{H}|_C = \{(h(x_1), ..., h(x_m)) : h \in \mathcal{H}\}$ contains the distinct binary behaviors of the class on C. The growth function is

$$\Pi_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X}: |C| = m} |\mathcal{H}|_C|. \tag{53}$$

The class shatters C exactly when this fixed restriction contains all 2^m binary vectors. Its VC dimension is the largest cardinality of a shattered set, or infinity if shattered sets have unbounded finite size.

To prove $\mathrm{VCdim}(\mathcal{H}) = d$, one constructs all labelings on one d-point set and proves that every $(d+1)$ -point set has a missing labeling. If $\mathrm{VCdim}(\mathcal{H}) \le d < \infty$, Sauer-Shelah states

$$\Pi_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i}. \tag{54}$$

Its proof removes one coordinate, splits prefixes into U and the double-extension family D, uses $|V| = |U| + |D|$ and $\mathrm{VCdim}(D) \le d - 1$, then closes the recurrence with Pascal’s identity.

Under binary zero-one loss and realizability, the resulting polynomial growth control supports the forward conclusion that finite VC dimension is distribution-free PAC learnable, with standard ERM upper-bound order $O\left(\frac{d \ln\left(\frac{1}{\varepsilon}\right) + \ln\left(\frac{1}{\delta}\right)}{\varepsilon}\right)$. Chapter 3 begins at the missing generalization machinery needed to remove realizability and prove the full agnostic equivalence; that machinery is not used in this reconstruction.

Key Takeaway

An infinite class can be statistically manageable because learning sees its finite-sample behaviors, not its raw cardinality. VC dimension identifies the largest set on which every binary behavior remains possible. Sauer-Shelah then turns a finite shattering limit into an exact polynomial growth bound. Together with the stated realizable VC result, this yields the Chapter 2 forward PAC guarantee while leaving the stronger agnostic equivalence for Chapter 3.

Optional self-study extension

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: Why do axis-aligned rectangles in $\mathbb{R}^2$ have VC dimension four?

Answer.

For the lower bound, place four points at the north, east, south, and west extremes of a cross. Given any subset of these four points, choose the smallest axis-aligned rectangle containing the selected points; when the subset is empty, place a rectangle away from all four. With the points in this position, an unselected extreme can be excluded by the corresponding side of the rectangle, so all 2^4 labelings are realized.

For the upper bound, take any five points. Choose a leftmost, rightmost, lowest, and highest point. These four roles name at most four distinct points, so at least one of the five points, call it q, is not needed as an extreme. Its two coordinates lie inside the coordinate ranges determined by the selected extreme points. Any axis-aligned rectangle containing all selected extremes must therefore contain q. Hence the labeling that makes the selected extremes positive and q negative is impossible. No five-point set is shattered, so the VC dimension is four.

Post-class or self-study. Not a prerequisite for the core path.

Source notes

The official Week 2 Learning Sheet v4 is used to preserve scope and sequence only: `w02-vc-dimension`, printed/PDF pp. 14-23; `w02-sauer`, pp. 24-31; and `w02-fundamental-preview`, pp. 32-35. The sheet's arbitrary-frequency sine shortcut is not used. Every shattering statement above has an explicit construction or a universal obstruction from the textbooks.

Definitions, claims, proofs, and examples were reconstructed from:

- Shalev-Shwartz and Ben-David, *Understanding Machine Learning*, Definitions 6.2-6.5 and Sections 6.2-6.3 (printed/PDF pp. 68-72), for restrictions, shattering, VC dimension, and the interval construction;
- Shalev-Shwartz and Ben-David, Definition 6.9 and Lemma 6.10, Section 6.5.1 (printed/PDF pp. 73-75), for the growth function, exact binomial bound, and projection proof;
- Shalev-Shwartz and Ben-David, Theorems 6.7-6.8 and Section 6.5 (printed/PDF pp. 72-78), for the binary realizable-PAC forward implication, its rate, and the boundary of the complete equivalence proof;
- Mohri, Rostamizadeh, and Talwalkar, *Foundations of Machine Learning*, 2nd ed., Sections 3.2-3.3 (printed pp. 34-37; PDF pp. 51-54), especially Figure 3.1, for growth-function notation and the interval VC calculation.

The optional rectangle calculation follows Shalev-Shwartz and Ben-David, Section 6.3.3 (printed/PDF p. 71). It has zero class minutes, exports no required knowledge, and is not needed by the core tutorial.